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ABSTRACT

A successful primary cementing operation is critical to maintaining the integrity of oil and gas wells. This operation includes injecting multi-fluid sequences into a narrow eccentric annulus, i.e., displacing the drilling mud first with a wash, followed by a spacer fluid, and finally, one or more cement slurries. We modify the two-dimensional gap averaged and the dispersive two-dimensional gap averaged models for multi-fluid displacement sequences. In particular, we implement a monotone numerical scheme, namely, the local Lax–Friedrichs scheme, for the evolution of the concentrations to ensure that individual fluid concentration bounds between $0 \leq \bar{c} \leq 1$, and the fluid concentrations sum to 1. The strong conservative properties of the numerical scheme are validated by achieving negligible relative errors (on the order of $\sim 10^{-15}$) for five distinct scenarios. The outcomes of this study can bring a significant improvement in designing the primary cementing operation, where an exact evaluation of fluid volumes is essential.

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I. INTRODUCTION

Seepage of oil and leakage of gas from hydrocarbon wells is an all too common occurrence, observable at onshore wells. In end-of-life and orphaned wells, oil seepage is often a sign of casing failure and/or pressure recovery in the reservoir. Gas leakage can occur due to a combination of cementing defects and operational factors: inadequate centralization, incomplete mud removal (channels or wet microannuli), cement shrinkage, production, completion, or geomechanical stresses leading to cracking or debonding. In each case, narrow pathways from a source to the surface are created along which gas may flow. Examples or reported leakage rates from Western Canada can be found in Refs. 1 and 2. Absence of similar studies from other gas-producing regions often reflects a lack of data management, poor public data access, or differing operational practices. For example, in Canada, surface vents are open, allowing leakage and also measurement. In contrast, in many US jurisdictions, surface vents are closed: pressure build-up often forces fugitive gas away from the wells, potentially into groundwater or to the surface over a wider area.

Consequently, well integrity and leakage have been a key concern of ethical producers for many decades. Here we focus on modeling the primary cementing process that leads to well integrity. In primary cementing, a sequence of fluids is pumped from the surface down

inside the casing to the bottom hole, returning upward in the annular space between the outer wall of the casing and the inner wall of the borehole (or previous casing) [see Fig. 1(a)]. From the fluid mechanics perspective, this is a displacement flow of a sequence of multiple fluids that takes place over distances $\sim 10^2 - 10^3$ m in length, involving an annular duct with typical gap sizes $\sim 0.1 - 5$ cm, due to eccentric placement of the casing in the hole. The hole is initially filled with a drilling fluid (mud), usually clay-based or polymeric, plus solids for weighting. Similar fluids (and water) are used to displace the drilling fluid, followed by the cement slurries, which are slowly reacting colloidal suspensions consisting of water, cement, and other additives. In general, these fluids have non-Newtonian rheologies, often modeled as Herschel–Bulkley fluids, and may have significantly different densities. Drilling mud density is determined by static and dynamic pressure requirements of drilling. Cement densities tend naturally to be much higher ($1600 - 1900$ kg/m³), although there are also lower density cements ($1400 - 1600$ kg/m³) and even ultralow special blends.

Given the extreme ratios of the well length to the annular gap thickness, the use of fully three-dimensional (3D) simulations in the design of primary cementing operations is restricted to the study of relatively short sections.^{3–6} Operational design usually uses a combination of 1D hydraulic simulations for pressure matching, preserving

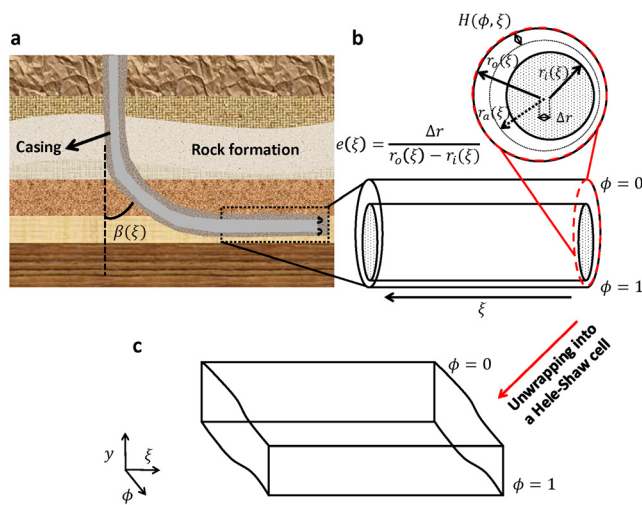


FIG. 1. (a) Well geometry, (b) schematic of eccentric annular space between the outer surface of casing and the rock formation, and (c) unwrapping the annulus to a Hele-Shaw cell with a non-uniform gap thickness.

primary well security, and 2D simulations that exploit the geometric aspect ratio, generally by a form of averaging across the annular gap. Examples illustrating the range of applications are readily found in Refs. 7–11. The underlying fluid mechanics model was introduced by Bittleston *et al.*¹² The model is developed based on a narrow gap assumption, reducing the Navier–Stokes equations to the 2D shear flow approximation, which is then simplified to a Hele–Shaw type flow: the 2D gap-averaged (2DGA) model [see Figs. 1(b) and 1(c)]. This model has been developed further in several ways: identifying steady state displacements;¹³ connecting with the 1D approaches;¹³ addressing horizontal annuli and moving casing;^{14,15} and modeling turbulent and mixed regimes.^{16,17} Although the 2DGA model is effective in predicting the overall flow behaviors, it neglects a wide range of interesting phenomena that might happen in the narrow gap space between casing and formation, e.g., formation of wall layers, and dispersion. This limitation has been recently addressed for Newtonian fluids^{18,19} by including an approximation of the dispersive behavior: the dispersive 2D gap-averaged approach (D2DGA). This is currently being extended to model non-Newtonian fluids.²⁰

A common feature of the above models is that they consist of a time-dependent transport equation for the individual fluids, physical closure laws that describe the assumed behavior of the fluids at the scale of the annular gap, plus an elliptic partial differential equation for the velocity field (or stream function). The latter is solved at each time-step, for given volume fractions (concentrations) of each fluid present locally, and an imposed flow rate. The transport problem is an advection-diffusion equation. In turbulent flows, Taylor dispersion can be dominant and results in anisotropic diffusion.¹⁶ For laminar regimes, molecular diffusion typically has a small effect, neglected in the D2DGA model.

The coupling of the fluid concentrations to the velocity computation takes place through closure expressions for the fluid rheology and density, which affect the local wall shear stress computations, and the effects of buoyancy on the flow. This means that the fluxes of each fluid are dependent on the fluid concentrations, which can lead to a

range of nonlinear behaviors for the conservation law. The challenge that we address in this paper is to find a computational algorithm that can cope with both the nonlinear behaviors and the vector of fluid concentrations. Regarding the nonlinear effects, a monotone scheme is needed to handle shocks and similar behaviors. These methods are relatively easy to find.

The vector of concentrations presents a different difficulty. Not only must each concentration (volume fraction) lie between 0 and 1, but also the concentrations must add up to 1. It is the latter constraint that is uncommon, i.e., summing to 1. In generic displacement flows with two fluids, one solves for a single concentration c (e.g., the volume of fluid method), assuming that $1 - c$ describes the remaining fluid. Such treatment may present a bias in the way in which the first fluid is treated. With a series of K fluids, ideally each fluid should evolve governed by an analogous equation. Bounding each fluid's concentration and ensuring the summation constraint must be a part of this system of conservation equations and hence the numerical solution. Adaptation and implementation of such a numerical scheme is the focus of this paper.

We have not found a numerical scheme that is readily designed for the specifics of our system. Instead, we have adapted the monotone methods of Zhang and Shu.²¹ The range of flow problems that are relevant to the cementing application is quite large, resulting in flows that are both complex and messy. From an industrial perspective, we aim to gain a physical understanding of unmeasured downhole phenomena so that the process design may be changed. Thus, excessively high order methods are not of direct interest compared to simplicity and robustness. Our main approach is based on a first order Lax–Friedrichs scheme, although later we make comments regarding a second order method.

Aside from the cementing application, there are other physical flows that are mathematically modeled in a similar way. First, multi-species porous media flows often also have a single velocity and pressure field, with a vector of fluid concentrations. Second, if we consider multi-layered flows in ducts (e.g., displacement/cleaning, manufacturing, co-extrusion, coating), we may have a series of thin film flow equations for each layer thickness, which must sum to the width of the duct. It is hoped that the methods developed in the present work can be simply adapted to these situations.

An outline of the paper is as follows: in Sec. II, we give a brief overview of how well cementing displacement flows have been modeled, using a reduced model. This includes the all important mass transport equations, the subject of the paper. The proposed numerical method is explained in Sec. III, focusing on the simplest gap-averaged models. Examples that illustrate the implementation for a range of practical flows are given in Sec. IV. We then move to more complex displacement flows in Sec. V, where we have to adapt the numerical method to significantly more complex fluxes, but are still able to ensure a stable monotone scheme. This is again illustrated with a range of examples. The paper ends with a discussion and summary.

II. MODELS OF CEMENTING DISPLACEMENT FLOW

Here we give a general overview of two-dimensional gap averaged (2DGA) cementing models, in terms of their structure and underlying assumptions, which lead to the transport problem associated with displacement flows, as addressed in this paper.

In general, these models are deduced from the Navier–Stokes equations by means of scaling assumptions that are explained in detail

elsewhere,^{12,16} but are similar to those involved in Hele–Shaw flows. Standard scaling arguments require a small annular gap ($\sim 2\hat{d}^*$) compared to both the length of the casing ($\hat{L}$) and the mean circumference ($2\pi\hat{r}_a^*$). The circumference is much smaller than the length, so $\hat{d}^*/(\pi\hat{r}_a^*) \ll 1$ is generally sufficient, together with geometry variations assumed to be slow with respect to $\hat{d}^*$, which is a representative difference between outer and inner radii. This suffices to simplify different gradients and eliminate curvature terms at leading order. For the momentum equations, we also require that $Re[\hat{d}^*/(\pi\hat{r}_a^*)] \ll 1$, for an appropriately defined gap-scaled Reynolds number Re , in order to eliminate relevant inertial terms.

The Navier–Stokes equations are reduced to a 2D shear flow. The y -coordinate measures across the annular gap: $y \in [-H, H]$, with coordinates (ϕ, ξ) in azimuthal and flow (axial) directions, respectively. The flows are incompressible, and the gap-averaged velocities in azimuthal and axial directions are denoted $(\bar{v}, \bar{w})$, which may be described in terms of a stream function Ψ :

$$\nabla_a \Psi = \left(\frac{1}{r_a} \frac{\partial \Psi}{\partial \phi}, \frac{\partial \Psi}{\partial \xi} \right) = (2H\bar{w}, -2H\bar{v}), \quad (1)$$

where r_a represents the local mean radius of the annulus. Here $\nabla_a = \left(\frac{1}{r_a} \frac{\partial}{\partial \phi}, \frac{\partial}{\partial \xi} \right)$ is the leading order 2D divergence operator.

The geometry is illustrated in Fig. 1. In practice, H can vary with ξ and ϕ , according to the irregularity of the outer wall, which may be either uniform (inside previous casing) or varying (drilled borehole). Additionally, the eccentricity (e) will vary slowly with ξ , depending on well inclination and casing centralizers' effectiveness. For simplicity, we assume that the axial derivatives of all geometric variables are small, i.e., the annulus is locally uniform. A typical description in a uniform annulus is

$$H(\phi) = 1 + e \cos \pi \phi, \quad (2)$$

for $e \in [0, 1)$ and $\phi \in [0, 1]$: $e = 0$ corresponds to a concentric annulus and $e = 1$ implies contact between the surface of the casing and the outer wall.

A. Momentum balance

The general approach now involves working with the reduced 2D shear flow, in directions (ϕ, ξ) . The pressure is constant across the annular gap. We usually make some assumptions regarding the distribution of fluids across the annular gap and then solve for the gap-averaged velocity $(\bar{v}, \bar{w})$ in terms of the imposed pressure gradient. This leads to an algebraic relationship between the pressure gradients and the stream function. By cross-differentiating, we can eliminate the pressure (or the stream function), leading to an equation of the form

$$\nabla_a \cdot [\mathbf{S} + \mathbf{b}] = 0, \quad (3)$$

where $\mathbf{S}$ and $\mathbf{b}$ represent the modified pressure gradient and buoyancy vector, respectively.

Equation (3) is generally a quasilinear elliptic equation for Ψ . In the case that the fluids have a yield stress, the function $\mathbf{S}$ might not be differentiable everywhere, and an elliptic variational inequality replaces (3). As examples, assuming the fluid concentrations are uniformly mixed across the annulus¹² allows one to deduce that the averaged

flow is in the direction of the modified pressure gradient, which leads to the 2DGA model that has been extensively studied. The variational inequality is outlined in Ref. 13. Assuming that the inner cylinder can either rotate or translate is dealt with by Carrasco-Teja and Frigaard,^{15,22} leading to a more complex version of (3). Moving away from the laminar flows,^{12,15} mixed and turbulent regimes are relevant in the field and have been modeled.¹⁶ The format of (3) does not change in any of these cases. More recently, the laminar flow assumption of a uniform fluid concentration across the gap has been improved by directly modeling dispersion on the scale of the annular gap.^{18,20} This leads to interesting effects of buoyancy and rheology that are not fully explored. Still, the model format remains as Eq. (3).

This paper is not concerned with solution of (3), but more directly with fluid transport. In practice, different methods can be used, including an asymptotic approximation,¹² full 2D solution using an augmented Lagrangian method to account for static zones,^{16,23} and lubrication-style approximations/simplifications to a near axial flow.^{14,24} In the current work, we simply assume that the stream function Ψ is calculable from (3), for a given set of fluid concentrations, at each timestep.

B. Transport of fluid species

The K different fluids sequentially pumped into the annulus are represented through a vector $\bar{\mathbf{c}}$ of concentrations, i.e., volume fraction, interpreted as the gap-averaged concentrations. In a typical cementing sequence, the *in situ* drilling mud ($k = 1$) is displaced by a wash ($k = 2$), then a spacer fluid, followed by one or more cement slurries, and finally, a drilling mud is used to displace the inside of the casing. Thus, $K = 5$ or more fluids is not uncommon. Given that the annulus is long and thin, if the displacement is effective, then the fluids only come into contact with the adjacent fluids. However, especially in an eccentric annulus, we may find that fluids channel through one another, due to adverse density or viscosity gradients. Thus, having more than two fluids present at any particular (ϕ, ξ) point in the annulus is a common occurrence.

To advance the fluid concentration $\bar{\mathbf{c}}_k$, we solve the following transport equation:

$$\frac{\partial [H r_a \bar{\mathbf{c}}_k]}{\partial t} + \nabla_a \cdot \mathbf{q}_k = 0, \quad (4)$$

where $\mathbf{q}_k$ is the volumetric flux of fluid k . The volumetric flux for fluid k has the following general form:²⁰

$$\mathbf{q}_k = r_a H(\bar{\mathbf{v}}, \bar{\mathbf{w}}) \mathbf{q}_{k,0} + b H^3 r_a I_{k,3} (-\sin \beta \sin \pi \phi, \cos \beta). \quad (5)$$

The functions $\mathbf{q}_{k,0}$ and $I_{k,3}$ will depend on $\bar{\mathbf{c}}$ and on the rheologies of the individual fluid components, plus the mean velocity (which drives the shear rate dependency). The parameter b is the buoyancy number. An example of these functions and parameters will be given later for the Newtonian D2DGA model.¹⁸ In general, we expect that

$$\sum_{k=1}^K \mathbf{q}_{k,0} = 1, \quad \sum_{k=1}^K I_{k,3} = 0 \quad (6)$$

to conserve the total volumetric flux.

For the 2DGA model, the description is greatly simplified as we find that $I_{k,3} = 0$ and $\mathbf{q}_{k,0} = \bar{\mathbf{c}}_k$, so that

$$\frac{\partial [Hr_a \bar{c}_k]}{\partial t} + \nabla_a \cdot [r_a H \bar{c}_k (\bar{v}, \bar{w})] = 0, \quad (7)$$

i.e., the fluid concentrations ($\bar{c}_k$) are advected with the gap-averaged velocities ($\bar{v}, \bar{w}$). However, note that the fluxes above are a nonlinear function of the concentrations, via the calculation of Ψ .

Three challenges arise in solving (7). First, sharp fronts may arise (shocks) due to the nonlinearity. Thus, the numerical scheme must calculate entropy solutions in a stable way. Second, because the concentrations represent volume fractions, we require that $\bar{c}_k \in [0, 1]$ for each k , at any (ϕ, ξ) and time t , which suggests a form of maximum principle be satisfied numerically. Third, we require that

$$\sum_{k=1}^K \bar{c}_k(\phi, \xi, t) = 1. \quad (8)$$

Finally, these properties should be satisfied not only for the simpler 2DGA transport equation but also for Eq. (4) with the most complex fluxes (5).

III. NUMERICAL METHOD

We use a finite volume method based on a staggered mesh (see Fig. 2). It is assumed that at timestep n , the concentrations $\bar{c}_{i+1/2,j+1/2}^n$ are known at the cell centers and can be used to define the fluid properties. Together with the imposed flow rate, this suffices to solve the elliptic equation (3). Thus, $\Psi = \Psi^n$ is known at the corners of the mesh (shown with a circle), typically calculated using a second order method. These discrete values of Ψ can be used to approximate $(\bar{v}, \bar{w})$ at the centers of the cell faces or edges (shown with a square).

Using a difference in time, the integral form of (7) is

$$\begin{aligned} [Hr_a \bar{c}_k]^{n+1} &= [Hr_a \bar{c}_k]^n \\ &+ \frac{\lambda}{2} \int_{\xi_j}^{\xi_{j+1}} \left[\bar{c}_k \frac{\partial \Psi}{\partial \xi} \right] (\phi_{i+1}) - \left[\bar{c}_k \frac{\partial \Psi}{\partial \xi} \right] (\phi_i) d\xi \\ &- \frac{\lambda}{2} \int_{\phi_i}^{\phi_{i+1}} \left[\bar{c}_k \frac{\partial \Psi}{\partial \phi} \right] (\xi_{j+1}) - \left[\bar{c}_k \frac{\partial \Psi}{\partial \phi} \right] (\xi_j) d\phi. \end{aligned} \quad (9)$$

The overbar in the first two terms denotes an area-averaged quantity, over a single mesh cell centered at $(\phi_{i+1/2}, \xi_{j+1/2})$. The parameter $\lambda = \Delta t / (\Delta \phi \Delta \xi)$. We have used the stream function in place of the velocity field to define the integral flux terms, which are evaluated along the faces of the mesh cell considered. This serves to better illustrate the global conservation property that results from incompressibility.

Let us assume that for each fluid k we use the same (consistent) method to approximate the concentrations $\bar{c}_k$ at the cell faces, and that $\sum_{k=1}^K \bar{c}_k = 1$ in this approximation, as would be expected. If we then sum (III) over all k , we see that the concentrations $\bar{c}_k$ sum to 1 in all flux terms, and the stream function terms then integrate to zero, leaving

$$\sum_{k=1}^K [Hr_a \bar{c}_k]^{n+1} = \sum_{k=1}^K [Hr_a \bar{c}_k]^n. \quad (10)$$

Since the geometry is time independent, we see that the concentrations at the new timestep $n+1$ will sum to 1, provided that they did so at timestep n .

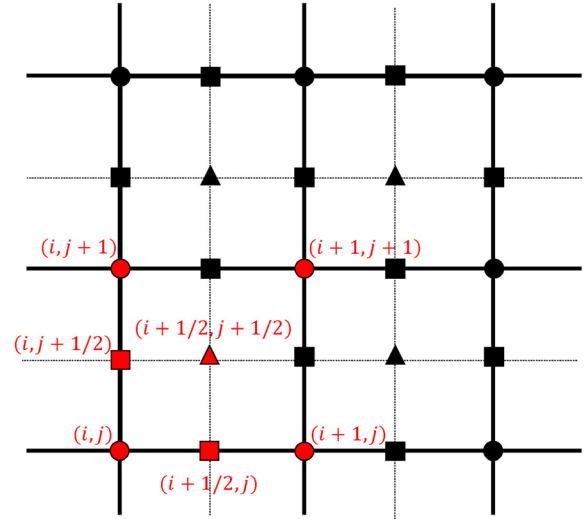


FIG. 2. Distribution of mesh points at cell centers, edges, and corners.

If instead of (7) we used (4) with the more complex fluxes (5), our integrals along the cell faces would contain the functions $q_{k,0}$ in place of $\bar{c}_k$, and also $I_{k,3}$. Again, provided that the approximations of these functions are consistent and are done in such a way that (6) is satisfied, then it is easy to see that the integral form of (4) will also conserve the sum of the $\bar{c}_k$ between timesteps. The fluxes involving $q_{k,0}$ sum to 1 again, to give integrals of only the stream function which integrate to zero. The fluxes involving $I_{k,3}$ sum to zero directly.

A. Monotone fluxes

Having outlined the underlying principles needed to ensure that the concentrations sum to 1 at each mesh point and timestep, we now proceed to specifics. The underlying method we use revolves around definition of monotone fluxes for the terms along the cell faces. We first illustrate the main idea using the simpler fluxes in (7). We approximate the concentrations in these integrals with a term evaluated at the midpoint, e.g.,

$$\int_{\xi_j}^{\xi_{j+1}} \left[\bar{c}_k \frac{\partial \Psi}{\partial \xi} \right] (\phi_i) d\xi \approx \bar{c}_k(\phi_i, \xi_{j+1/2}) [\Psi_{i,j+1} - \Psi_{i,j}].$$

We define monotone fluxes using a Lax–Friedrichs expression for each as

$$h_{\phi,i,j+1/2}(\bar{c}_k^+, \bar{c}_k^-) = \frac{(\Psi_{i,j} - \Psi_{i,j+1})}{2} (\bar{c}_k^+ + \bar{c}_k^-) - a_{\phi,i,j+1/2} (\bar{c}_k^+ - \bar{c}_k^-), \quad (11)$$

$$h_{\xi,i+1/2,j}(\bar{c}_k^+, \bar{c}_k^-) = \frac{(\Psi_{i+1,j} - \Psi_{i,j})}{2} (\bar{c}_k^+ + \bar{c}_k^-) - a_{\xi,i+1/2,j} (\bar{c}_k^+ - \bar{c}_k^-). \quad (12)$$

In each case, $\bar{c}_k^+$ and $\bar{c}_k^-$ represent approximations to $\bar{c}_k^n$ on the positive and negative sided of the cell face, e.g., in (11), using a first order approximation, we have

$$\bar{c}_k^+ = \bar{c}_{k,i+1/2,j+1/2}^n, \quad \bar{c}_k^- = \bar{c}_{k,i-1/2,j+1/2}^n.$$

The terms $a_{\phi,i,j+1/2}$ and $a_{\xi,i+1/2,j}$ are

$$a_{\phi,i,j+1/2} = \frac{|\Psi_{i,j} - \Psi_{i,j+1}|}{2}, \quad a_{\xi,i+1/2,j} = \frac{|\Psi_{i+1,j} - \Psi_{i,j}|}{2}. \quad (13)$$

We note that with this choice of $a_{\phi,i,j+1/2}$ and $a_{\xi,i+1/2,j}$, we effectively have a local Lax–Friedrichs flux (LLF), reasons for which will be illustrated later.

B. Monotonicity of $\bar{c}_k$

The mesh function $\bar{c}_{k,i+1/2,j+1/2}^{n+1}$ is computed from

$$\begin{aligned} (\bar{H}\bar{r}_a)_{i+1/2,j+1/2} [\bar{c}_{k,i+1/2,j+1/2}^{n+1} - \bar{c}_{k,i+1/2,j+1/2}^n] \\ = -\frac{\lambda}{2} [h_{\phi,i+1/2,j+1/2}(\bar{c}_k^+, \bar{c}_k^-) - h_{\phi,i,j+1/2}(\bar{c}_k^+, \bar{c}_k^-)] \\ -\frac{\lambda}{2} [h_{\xi,i+1/2,j+1/2}(\bar{c}_k^+, \bar{c}_k^-) - h_{\xi,i+1/2,j}(\bar{c}_k^+, \bar{c}_k^-)]. \end{aligned} \quad (14)$$

The difference formula for $\bar{c}_{k,i+1/2,j+1/2}^{n+1}$ above depends on $\bar{c}_{i+1/2,j+1/2}^n$ and its four neighboring meshpoints. The coefficients are

$$\begin{aligned} \bar{c}_{k,i+1/2,j+1/2}^{n+1} = \alpha_C \bar{c}_{k,i+1/2,j+1/2}^n + \alpha_N \bar{c}_{k,i+1/2,j+3/2}^n \\ + \alpha_E \bar{c}_{k,i+3/2,j+1/2}^n + \alpha_S \bar{c}_{k,i+1/2,j-1/2}^n \\ + \alpha_W \bar{c}_{k,i-1/2,j+1/2}^n, \end{aligned} \quad (15)$$

$$\alpha_C = 1 - \frac{\lambda [a_{\phi,i,j+1/2} + a_{\phi,i+1,j+1/2} + a_{\xi,i+1/2,j} + a_{\xi,i+1/2,j+1}]}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}}, \quad (16)$$

$$\alpha_N = \frac{\lambda [a_{\xi,i+1/2,j+1} - \frac{(\Psi_{i+1,j+1} - \Psi_{i,j+1})}{2}]}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}}, \quad (17)$$

$$\alpha_E = \frac{\lambda [a_{\phi,i+1,j+1/2} - \frac{(\Psi_{i+1,j+1} - \Psi_{i+1,j})}{2}]}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}}, \quad (18)$$

$$\alpha_S = \frac{\lambda [a_{\xi,i+1/2,j} + \frac{(\Psi_{i+1,j} - \Psi_{i,j})}{2}]}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}}, \quad (19)$$

$$\alpha_W = \frac{\lambda [a_{\phi,i,j+1/2} + \frac{(\Psi_{i,j+1} - \Psi_{i,j})}{2}]}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}}. \quad (20)$$

We observe that each of the coefficients α_N , α_E , α_S , and α_W are positive and that $\alpha_C \geq 0$, provided that

$$\Delta t \leq \Delta \phi \Delta \xi \min_{i,j} \Lambda_{i,j}, \quad \text{where}$$

$$\Lambda_{i,j} = \frac{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}}{a_{\phi,i,j+1/2} + a_{\phi,i+1,j+1/2} + a_{\xi,i+1/2,j} + a_{\xi,i+1/2,j+1}}. \quad (21)$$

Note that (21) is a form of CFL condition, as the denominator above consists of differences in stream function values at the corner points. These are effectively $2H \times \bar{v}$ or $\bar{w}$, multiplied by either mesh spacing. Since the speed decreases where the annular gap width H is narrow, we see that using the LLF fluxes allow (21) to adjust sensibly to the

geometry. The actual timestep (Δt) is obtained by setting equality and multiplying the right-hand side of (21) by a CFL number < 1 .

Considering the coefficients above and provided that (21) holds, we can see the following:

1. Since $\alpha_N + \alpha_E + \alpha_S + \alpha_W + \alpha_C = 1$, we see that $\bar{c}_{k,i+1/2,j+1/2}^{n+1}$ is a convex combination of the $5\bar{c}_k^n$ values in (15), at the previous timestep.
2. The right hand side of (15) increases monotonically with any of the $5\bar{c}_k^n$ values.
3. If all of the $5\bar{c}_k^n = 0$, we have that $\bar{c}_{k,i+1/2,j+1/2}^{n+1} = 0$.
4. If all of the $5\bar{c}_k^n = 1$, we have $\bar{c}_{k,i+1/2,j+1/2}^{n+1} = 1$.

It follows that, provided that $\bar{c}_{k,i+1/2,j+1/2}^n \in [0, 1]$ for all i, j , then $\bar{c}_{k,i+1/2,j+1/2}^{n+1} \in [0, 1]$ for all i, j (see, e.g., Zhang and Shu²¹). Furthermore, we have already shown that at any mesh point, the concentrations summed over k will add up to 1 at timestep $n + 1$, provided this was true at step n .

IV. 2DGA MODEL EXAMPLES

To illustrate the conservation properties of the numerical scheme, we present results from five sample cases using the 2DGA model. In each example, the well is 200 m long and is meshed with 400 meshpoints axially and 20 meshpoints azimuthally. The inner and outer diameters of the annulus are 0.254 and 0.355 m, respectively. As the aim of the paper is to explore multi-species displacement flows, in each case three fluids are pumped in, as indicated in Table I, and the fluids are pumped at a constant flow rate (here $\dot{Q}_0 = 0.004865 \text{ m}^3/\text{s}$). In each case, we fix $n_1 = n_2 = n_3 = 1$.

Cases 1–3 use Newtonian fluids. In case 1, the annulus is vertical with eccentricity $e = 0.2$. Maintaining a density and viscosity hierarchy between displaced and displacing fluids in a sequence is known to result in stable steady displacements for vertical wells. Figure 3 shows color maps of the three fluid concentrations together with streamlines of the flow, at successive times throughout the displacement. The black lines are plotted at $\bar{c}_1 = 0.9$, $\bar{c}_1 = 0.5$, $\bar{c}_1 = 0.1$, to illustrate the spreading of the displacement front. We see that the initial fluid 1 to fluid 2 front is relatively sharp, but spreads axially. The positive density and viscosity difference counter this spreading behavior and the effect

TABLE I. Overview of parameter ranges in different case study scenarios. Power law indices selected are $n_1 = n_2 = n_3 = 1$.

Parameters	Case 1	Case 2	Case 3	Case 4	Case 5
$\hat{\rho}_1 (\text{kg/m}^3)$	1200	1200	1200	1200	1200
$\hat{\kappa}_1 (\text{Pa s})$	0.05	0.05	0.05	0.05	0.05
$\hat{\tau}_{1,Y} (\text{Pa})$	0	0	0	8	0
$\hat{\rho}_2 (\text{kg/m}^3)$	1300	1300	1192	1300	1210
$\hat{\kappa}_2 (\text{Pa s})$	0.075	0.75	0.075	0.75	0.05
$\hat{\tau}_{2,Y} (\text{Pa})$	0	0	0	0	0
$\hat{\rho}_3 (\text{kg/m}^3)$	1400	1400	1400	1400	1220
$\hat{\kappa}_3 (\text{Pa s})$	0.1	0.1	0.1	0.1	0.05
$\hat{\tau}_{3,Y} (\text{Pa})$	0	0	0	5	0
β	0	90	0	(0 – 90)	90
e	0.2	0.8	0.01	(0.2 – 0.8)	0

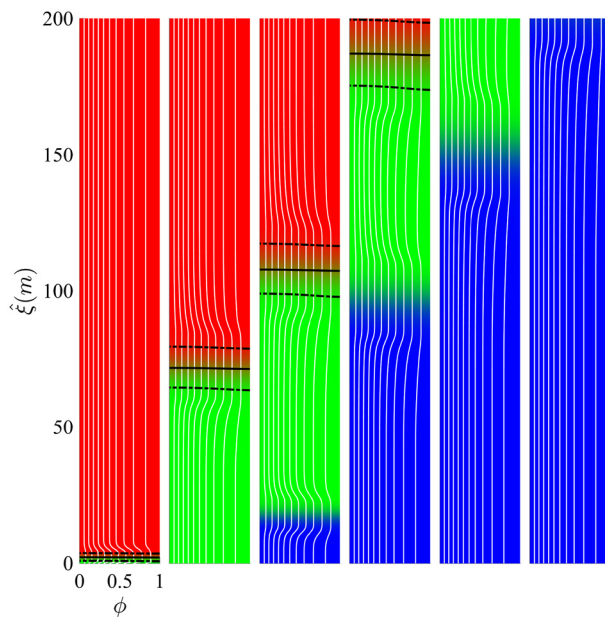


FIG. 3. 2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9,$ and 2987.5 s are shown for case 1. White lines represent streamlines and black lines correspond to $\bar{c}_2 = 0.1, 0.5,$ and 0.9 from top to bottom. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

of eccentricity. The viscosity and density differences are reduced when considering the fluid 2 to fluid 3 front, and the frontal region extends further axially than the previous interface. The spreading behavior is likely due to a combination of numerical diffusion from the first order method and physical dispersion, which drives azimuthal secondary flows in the case of steady displacements.²³

Case 2 uses a similar fluid sequence, but now the annulus is horizontal and strongly eccentric $e = 0.8$. Although steady displacements are possible in a horizontal annulus,¹⁴ more commonly there is a competition between the tendency of the fluids to slump under gravity and the geometric effects of eccentricity. This is the case here. The large eccentricity causes the fluids to flow more rapidly along the upper (wide, $\phi = 0$) side of the annulus; observe the densely packed streamlines. However, as the front advances, due to the density difference, we see buoyancy driven fingering emerging as the heavier fluid 2 attempts to push azimuthally down into fluid 1 along the narrow side. The azimuthal flow is viscosity stable but density unstable. The same phenomena are seen as fluid 3 displaces fluid 2.

For case 3, we have selected a density unstable situation in a very mildly eccentric vertical annulus. Despite the small density difference between fluids 1 and 2, we see that fluid 2 displaces rapidly along the wide side of the annulus. Such behavior is common where washes are used (and likely more extreme due to larger density differences). In this example, fluid 3 reverts to the stable density and viscosity hierarchy and displaces the fluids ahead, as illustrated.

In case 4, we illustrate a more complex displacement in which the fluid sequence is non-Newtonian, including a yield stress. The annulus is also no longer uniform. Instead, we suppose that the inclination of the well changes from vertical to horizontal with depth. As the inclination changes, we suppose that the eccentricity increases, i.e., the steel

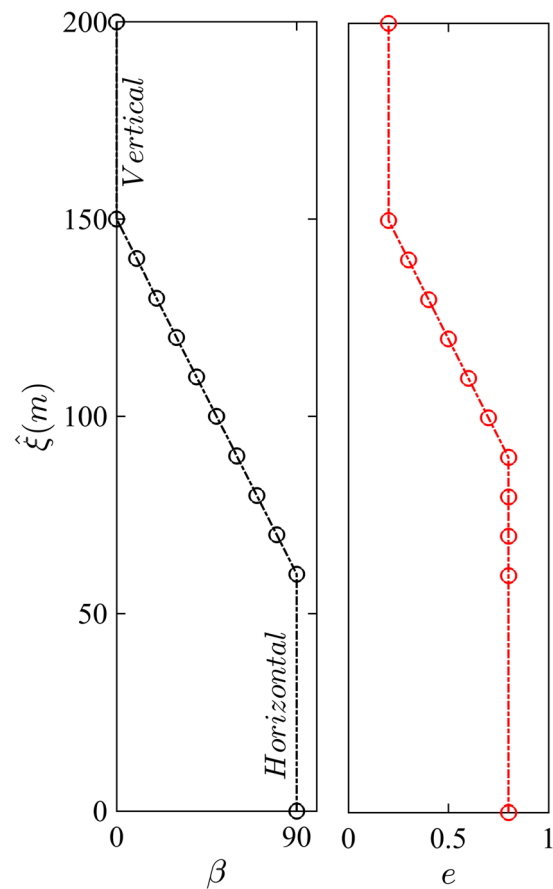


FIG. 4. Case 4: variation of inclination angle (β) and eccentricity (e) along the measured depth.

casing is positioned lower as the annulus becomes increasingly horizontal. The assumed profile of β and e is illustrated in Fig. 4.

The displacement flow is illustrated in Fig. 5. The initial flow of fluid 1 (red) shows a significant region of stationary fluid toward the narrow side of the annulus. The pressure gradient is insufficient to overcome the yield stress of fluid 1. As (the green) fluid 2 enters, it too moves along the wide side of the annulus. Fluid 2 has no yield stress but is relatively viscous. In the horizontal section, the buoyancy gradient due to the density difference cannot help. The pressure gradient exerted on fluid 1 is solely that generated by the viscous fluid 2 passing through the flowing area on the wide side of the annulus. This is insufficient to mobilize fluid 1 in the horizontal section of the well, where $e = 0.8$. We see that as the inclination moves away from horizontal (after 60 m), the channel of displacing fluid widens. This coincides with the increase in axial buoyancy force helping the displacement. Eventually, as also the eccentricity is reduced (Fig. 4), fluid 2 spreads across to the narrow side and displaces the full annulus. Fluid 3 (blue) enters the annulus at 900 s. We see the emergence of azimuthal density driven fingering instabilities as the heavier fluid 3 moves over the less dense fluid 1 layers, displacing fluid 2. These azimuthal instabilities reduce on entering the more vertical segment of the well. A substantial part of the well retains fluid 1 (drilling mud) at the end of the operation.

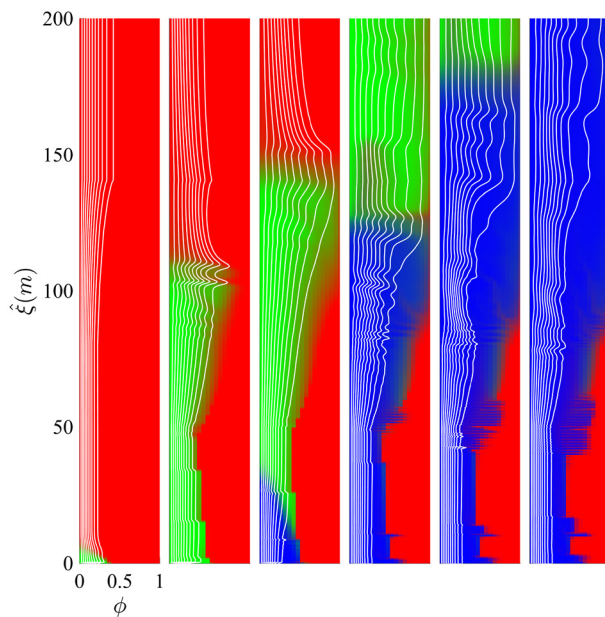


FIG. 5. 2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9,$ and 2987.5 s are shown for case 4. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

Figure 6 illustrates a simple example of buoyancy driven slumping flows in horizontal wells. The annulus is concentric, the fluids have the same viscosity and the slight density difference between the fluids leads to mild slumping displacement flow toward the lower side of the

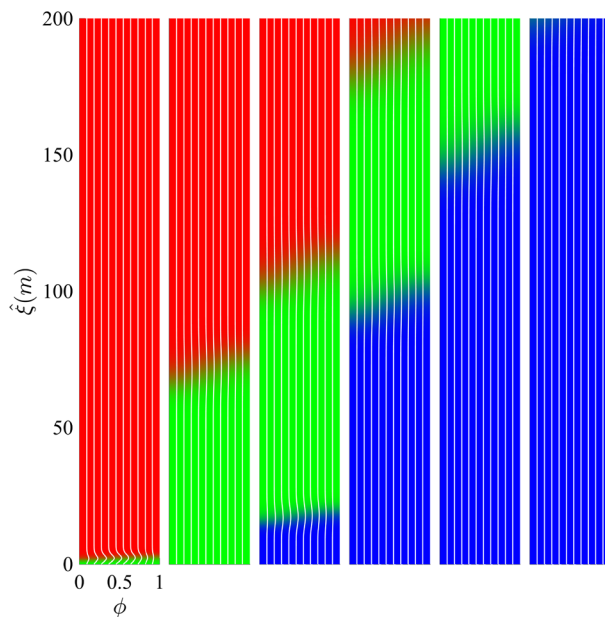


FIG. 6. 2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9,$ and 2987.5 s are shown for case 5. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

annulus ($\phi = 1$). At the same time, we see numerical diffusion effects in the axial direction, similar to those of case 1.

To verify the conservation properties of the numerical scheme, for each fluid, we compute the total volume in the annulus in two ways. First, we compute $Vol_1(k)$, by integrating $\bar{c}_{k,i+1/2,j+1/2}^n$ numerically over the interior volume of the annulus, i.e., we multiply by $Hr_a \Delta \phi \Delta \xi$ and sum over the interior cells. Second, we compute $Vol_2(k)$ at each timestep, starting with the initial volumes in the annulus and adding/subtracting the inflow/outflow of each fluid at the start and finish of the annulus, at each timestep. We normalize these quantities with the total annulus volume and subtracted to give a relative volumetric error. This is shown in Fig. 7, for cases 3–5. The relative errors are color-coded, corresponding to the colors used in the displacement figures. The errors rise and fall with the approximate volumes of fluids in the annulus, but remain consistently $O(10^{-15})$, comparable to the standard Matlab precision.

Similar errors are found for cases 1 and 2. Despite the increased physical complexity of flows 3 and 4, we see that the conservation properties are excellent. Fundamentally, the mass conservation problem is the same in each of these flows. By this, we mean that the volume fluxes are proportional to the mean velocity and concentration locally. Geometric complexity and the inclusion of non-Newtonian fluid properties may affect the speed of computation of the stream function (or potentially its accuracy). However, provided that the stream function values are fixed at each side of the annulus, evolving the mass conservation equations and conserving volumes is insensitive to the flow complexity.

V. DISPERSIVE AND DIFFUSIVE DISPLACEMENTS: THE D2DGA MODEL

The D2DGA model was introduced in Ref. 18 in order to redress the assumptions of earlier models,^{12,16} that assumed that the fluid

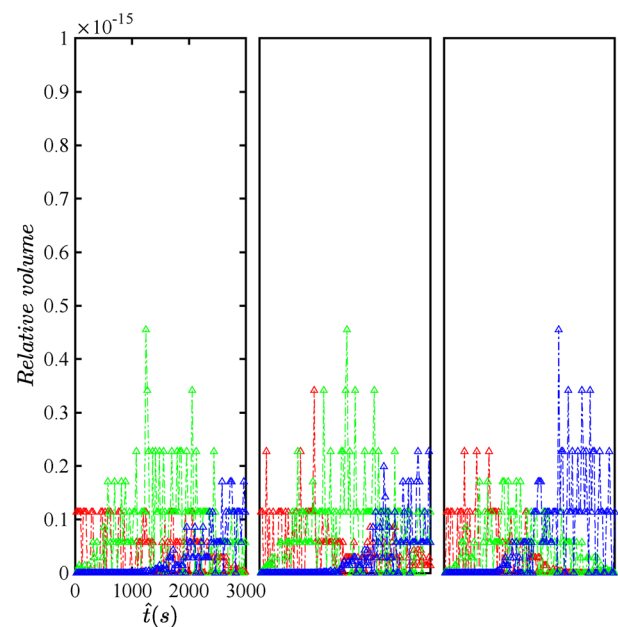


FIG. 7. 2DGA model: Conservation errors of fluid 1 (red), fluid 2 (green), and fluid 3 (blue) for cases 3–5 (from left to right).

concentrations were uniformly mixed across the annular gap. While reasonable for turbulent flows, laminar flows will disperse on the scale of the annular gap. The D2DGA model is built on an idealized process in which fluid 2 displaces fluid 1, dispersing in a finger along the center of the channel (representing the local narrow gap of the annulus). The resulting layered flow (see Fig. 8) to leading order is a shear flow, which enables one to solve for the velocity components and eventually compute the volumetric fluxes. This has been derived analytically for two Newtonian fluids¹⁸ and extended (mostly computationally) to displacement flows involving pairs of Herschel–Bulkley fluids.²⁰

The advantage of the approach developed is that the underlying scaling assumptions and reduction to a Hele–Shaw shear flow proceed as for the 2DGA model, leading to an elliptic equation (3) that represents the reduced order momentum balance. The specific expressions for $\mathbf{S}$ and $\mathbf{b}$ are given elsewhere.^{18,20} Here we simply assume that (3) is solved to give the stream function Ψ at each timestep. Our concern is with mass transport, which is governed by (4), with fluxes that satisfy the conservation conditions (6). Supposing we have two Newtonian fluids, then^{18,20}

$$q_{2,0}(\bar{c}; m) = \bar{c} \left[\frac{m\bar{c}^2 + 1.5(1 - \bar{c}^2)}{m\bar{c}^3 + 1 - \bar{c}^3} \right], \quad \bar{c} = \bar{c}_2, \quad (22)$$

$$I_{2,3}(\bar{c}; m) = \frac{\bar{c}^2(1 - \bar{c})^3 [4m\bar{c} + 3(1 - \bar{c})]}{12[m\bar{c}^3 + 1 - \bar{c}^3]}. \quad (23)$$

Here c is the concentration of the displacing fluid 2. The (scalar) buoyancy parameter is

$$b = -\frac{\Delta\hat{\rho}\hat{g}\hat{d}^{*2}}{r_a(\hat{\mu}_1\hat{\mu}_2)^{1/2}\hat{w}_0}, \quad (24)$$

with $\Delta\hat{\rho} = \hat{\rho}_1 - \hat{\rho}_2$ and $m = \hat{\mu}_1/\hat{\mu}_2$. We then simply define $q_{1,0} = 1 - q_{2,0}$ and $I_{1,3} = -I_{2,3}$, in order to satisfy (6).

A. Multifluid sequences

A K -fluid displacement sequence requires some care in modeling. As far as we know, there are very few studies of multi-fluid

displacement flows. In the cementing context, although there are multiple fluids pumped sequentially, the volumes are large, typically extending many hundreds of hole diameters axially. This is simply because the usual intention of the individual fluids is to separate from others in the sequence. Nevertheless, one needs a closure model to define the fluxes of each fluid, when > 2 are present locally.

Cementing flows are high Péclet number miscible displacements, viewed on the (2–3 cm) length-scale of the annular gap width. This means that on the initial displacement of a few meters, a well-defined front will remain, but on longer timescales as this front propagates along the well, residual fluid either mixes diffusively across the thinning wall layers, or may mix due to shear instability or other unsteadiness. Generalizing Fig. 8 to multiple layers is unlikely to reflect physical reality over many tens of meters and would also be prohibitively complex algebraically. Instead, the new fluid in the sequence (say fluid 3), initially only encounters the preceding fluid (i.e., fluid 2). When eventually the fluid 3 breaks through fluid 2, we assume that any small residual of fluid 1 is effectively mixed with fluid 2.

The above scenario allows us to adapt the two-fluid D2DGA model to the more complex physical situation. In our case, this is done by designating as the *displacing fluid* ($k = kd$), the newest fluid that has any significant presence. The other fluids present at the location ($k < kd$) are treated as a mixture of displaced fluids. Initially, the domain is filled with fluid 1 and therefore $kd = 2$. Thereafter, index kd can be identified locally by searching for the largest $k \geq 2$ for which $\bar{c}_k > 0$ (or above a predefined small tolerance concentration). The two-fluid definitions are now changed as follows:

$$q_{kd,0}(\bar{c}; m) = \bar{c}_{kd} \left[\frac{m\bar{c}_{kd}^2 + 1.5(1 - \bar{c}_{kd}^2)}{m\bar{c}_{kd}^3 + 1 - \bar{c}_{kd}^3} \right], \quad (25)$$

$$q_{k,0}(\bar{c}; m) = [1 - q_{kd,0}(\bar{c}; m)] \frac{\bar{c}_k}{1 - \bar{c}_{kd}}, \quad k \neq kd, \quad (26)$$

$$I_{kd,3}(\bar{c}; m) = \frac{\bar{c}_{kd}^2(1 - \bar{c}_{kd})^3 [4m\bar{c}_{kd} + 3(1 - \bar{c}_{kd})]}{12[m\bar{c}_{kd}^3 + 1 - \bar{c}_{kd}^3]}, \quad (27)$$

$$I_{k,3}(\bar{c}; m) = -I_{kd,3}(\bar{c}; m) \frac{\bar{c}_k}{1 - \bar{c}_{kd}}, \quad k \neq kd. \quad (28)$$

In Eqs. (26) and (28), the viscosity ratio m must be defined by replacing index 2 with kd and using some form of interpolant for the displaced fluid properties. Here, we extend the effective viscosity defined in Ref. 20 for multi-fluid sequences. In particular, the viscosity ratio is defined as: $m = (\frac{\hat{\mu}_e}{\hat{\mu}_{kd}})^K$ where $\hat{\mu}_e = (\prod_{k=1}^K \hat{\mu}_k)^{1/K}$. Equally the buoyancy parameter b in (24) needs to be adjusted so that $\Delta\hat{\rho}$ reflects the difference between displaced and displacing fluid densities. The density of a mixture of displaced fluids is obtained using a simple volume average. All of this is defined locally at each computational cell.

The above closure is what we have used in the examples presented below. It is not the only possible closure. At minimum, the constraints (6) must be satisfied, and the closure models must default to the underlying two-fluid expressions if $K = 2$.

B. Diffusive fluxes

In the mass transport problem of the 2DGA models,^{12,16} there is often a diffusive flux also included. For example, in laminar flows, one can include molecular diffusivity effects. For turbulent flows,¹⁶ Taylor

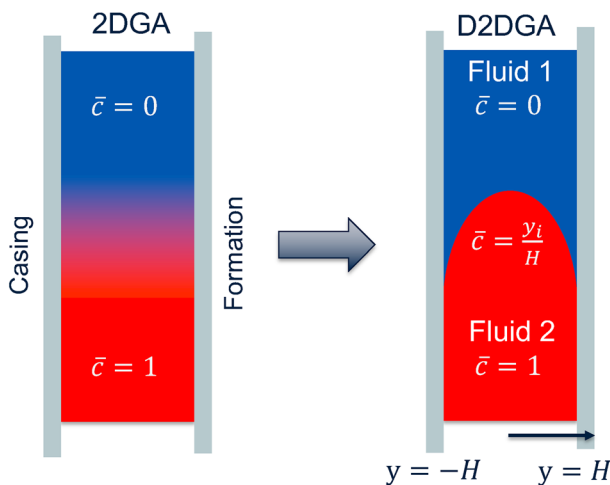


FIG. 8. Schematic of 2DGA and D2DGA models.

dispersion in the flow direction is the dominant effect, followed by turbulent diffusivity. The general form for including these effects is by adding a term to (4):

$$\frac{\partial[Hr_a \bar{c}_k]}{\partial t} + \nabla_a \cdot \mathbf{q}_k = \nabla_a \cdot [Hr_a D_d \nabla_a \bar{c}_k], \quad (29)$$

where D_d is a dimensionless diffusivity that will depend on the fluid concentrations and properties. In most cases, it is found that $D_d \ll 1$, as most cementing flows are advection dominated. In laminar flows in particular, molecular diffusivity is insignificant, which has been the motivation for instead looking at dispersion as the main mechanism for spreading of the displacement front. Even for turbulent Taylor dispersion,¹⁶ the physical effects are small enough that using explicit time-stepping for the diffusive terms does not typically introduce timestep restrictions more severe than the CFL conditions.

From the numerical perspective explored in this paper, we just note that since the fluid species are diffusing within one another, the net effect of diffusion should be zero. This is satisfied if the diffusivity D_d is the same for each fluid, since then $\sum_{k=1}^K D_d \nabla_a \bar{c}_k = (0, 0)$. The discretization used then mimics the treatment of $I_{k,3}$ outlined below. In general, we would evaluate D_d using the concentrations on a cell face (averaged from each side) and then simply differentiate the concentrations on either side using $\bar{c}_k^+$ and $\bar{c}_k^-$. The Taylor dispersion terms are slightly different in that they involve anisotropic diffusion, in the flow direction. However, these still fit the same framework.

C. Numerical approximation

The numerical approximation using the LLF method follows that outlined previously. In place of (11) and (12), we have

$$\begin{aligned} h_{\phi,i,j+1/2}(\bar{c}_k^+, \bar{c}_k^-) &= \frac{(\Psi_{ij} - \Psi_{i,j+1})}{2} (q_{k,0}(\bar{c}^+) + q_{k,0}(\bar{c}^-)) \\ &\quad - a_{\phi,q,i,j+1/2}(\bar{c}_k^+ - \bar{c}_k^-) \\ &\quad - b[r_a H^3 \sin \beta \sin \pi \phi]_{i+1/2,j+1/2} I_{k,3}(\bar{c}^+) \\ &\quad - b[r_a H^3 \sin \beta \sin \pi \phi]_{i-1/2,j+1/2} I_{k,3}(\bar{c}^-) \\ &\quad - a_{\phi,b,i,j+1/2}(\bar{c}_k^+ - \bar{c}_k^-), \end{aligned} \quad (30)$$

$$\begin{aligned} h_{\xi,i+1/2,j}(\bar{c}_k^+, \bar{c}_k^-) &= \frac{(\Psi_{i+1,j} - \Psi_{ij})}{2} ((q_{k,0}(\bar{c}^+) + q_{k,0}(\bar{c}^-)) \\ &\quad - a_{\xi,q,i+1/2,j}(\bar{c}_k^+ - \bar{c}_k^-) \\ &\quad + b[r_a H^3 \cos \beta]_{i+1/2,j+1/2} I_{k,3}(\bar{c}^+) \\ &\quad + b[r_a H^3 \cos \beta]_{i+1/2,j-1/2} I_{k,3}(\bar{c}^-) \\ &\quad - a_{\xi,b,i+1/2,j}(\bar{c}_k^+ - \bar{c}_k^-)). \end{aligned} \quad (31)$$

Again $\bar{c}_k^+$ and $\bar{c}_k^-$ represent approximations to $\bar{c}_k^n$ on the positive and negative sided of the relevant cell face. The $a_{\phi,q,i,j+1/2}$ and $a_{\xi,q,i+1/2,j}$ are selected to ensure monotonicity of the fluxes, i.e.,

$$\begin{aligned} a_{\phi,q,i,j+1/2} &= \frac{|\Psi_{ij} - \Psi_{i,j+1}|}{2} \\ &\quad \times \max_k \left\{ \left| \frac{\partial q_{k,0}}{\partial \bar{c}_k^+}(\bar{c}^+) \right|, \left| \frac{\partial q_{k,0}}{\partial \bar{c}_k^-}(\bar{c}^-) \right| \right\}, \end{aligned} \quad (32)$$

$$\begin{aligned} a_{\xi,q,i,j+1/2} &= \frac{|\Psi_{i+1,j} - \Psi_{ij}|}{2} \\ &\quad \times \max_k \left\{ \left| \frac{\partial q_{k,0}}{\partial \bar{c}_k^+}(\bar{c}^+) \right|, \left| \frac{\partial q_{k,0}}{\partial \bar{c}_k^-}(\bar{c}^-) \right| \right\}. \end{aligned} \quad (33)$$

The terms $a_{\phi,b,i,j+1/2}$ and $a_{\xi,b,i+1/2,j}$ are

$$\begin{aligned} a_{\phi,b,i,j+1/2} &= \max_k \left\{ |b(r_a \sin \beta)_{j+1/2} [H^3 \sin \pi \phi]_{i+1/2,j+1/2}|, \right. \\ &\quad \times |b(r_a \sin \beta)_{j+1/2} [H^3 \sin \pi \phi]_{i-1/2,j+1/2}| \\ &\quad \times \max_k \left\{ \left| \frac{\partial I_{k,3}}{\partial \bar{c}_k^+}(\bar{c}^+) \right|, \left| \frac{\partial I_{k,3}}{\partial \bar{c}_k^-}(\bar{c}^-) \right| \right\}, \end{aligned} \quad (34)$$

$$\begin{aligned} a_{\xi,b,i,j+1/2} &= \max_k \left\{ |[br_a H^3 \cos \beta]_{i+1/2,j+1/2}|, \right. \\ &\quad \times |[br_a H^3 \cos \beta]_{i+1/2,j-1/2}| \\ &\quad \times \max_k \left\{ \left| \frac{\partial I_{k,3}}{\partial \bar{c}_k^+}(\bar{c}^+) \right|, \left| \frac{\partial I_{k,3}}{\partial \bar{c}_k^-}(\bar{c}^-) \right| \right\}. \end{aligned} \quad (35)$$

To evaluate these derivative terms, note that effectively $q_{k,0}$ and $I_{k,3}$ are defined via the displacing fluid concentration $\bar{c}_{kd}$ in (26) and (28). Writing $\bar{c}'_{kd} = 1 - \bar{c}_{kd}$, we find that for $k \neq kd$:

$$\frac{\partial q_{k,0}}{\partial \bar{c}_k} = \frac{\bar{c}_k}{\bar{c}'_{kd}} \frac{\partial q_{kd,0}}{\partial \bar{c}_{kd}} + \frac{1 - q_{kd,0}}{\bar{c}'_{kd}} \left[1 - \frac{\bar{c}_k}{\bar{c}'_{kd}} \right], \quad (36)$$

which we recognize as interpolating between $[1 - q_{kd,0}]/\bar{c}'_{kd}$ and $\frac{\partial q_{kd,0}}{\partial \bar{c}_{kd}}$. Similarly for $I_{k,3}$.

Therefore, in evaluating the derivatives in the expressions for $a_{\phi,q,i,j+1/2}$ and $a_{\xi,q,i+1/2,j}$, we only need to differentiate the function $q_{kd,0}(\bar{c}; m)$ with respect to $\bar{c}_{kd}$ and may then bound above these expressions, i.e., we may use the right hand side of

$$\begin{aligned} a_{\phi,q,i,j+1/2} &\leq \frac{|\Psi_{ij} - \Psi_{i,j+1}|}{2} \\ &\quad \times \max \left\{ \left| \frac{\partial q_{kd,0}}{\partial \bar{c}_{kd}^+}(\bar{c}^+) \right|, \frac{|1 - q_{kd,0}(\bar{c}^+)|}{1 - \bar{c}_{kd}^+}, \left| \frac{\partial q_{kd,0}}{\partial \bar{c}_{kd}^-}(\bar{c}^-) \right|, \frac{|1 - q_{kd,0}(\bar{c}^-)|}{1 - \bar{c}_{kd}^-} \right\} \end{aligned} \quad (37)$$

and make similar adjustments to conservatively evaluate $a_{\xi,q,i,j+1/2}$, and the derivatives of $I_{k,3}$ in $a_{\phi,b,i,j+1/2}$ and $a_{\xi,b,i+1/2,j}$.

D. Monotonicity of the fluxes

We again use the cell-centered $\bar{c}_{i+1/2,j+1/2}^n$ to approximate the $\bar{c}^{\pm}$ at first order. The difference scheme becomes

$$\begin{aligned} \bar{c}_{k,i+1/2,j+1/2}^{n+1} &= F_{k,C}(\bar{c}_{k,i+1/2,j+1/2}^n) + F_{k,N}(\bar{c}_{k,i+1/2,j+3/2}^n) \\ &\quad + F_{k,E}(\bar{c}_{k,i+3/2,j+1/2}^n) + F_{k,S}(\bar{c}_{k,i+1/2,j-1/2}^n) \\ &\quad + F_{k,W}(\bar{c}_{k,i-1/2,j+1/2}^n), \end{aligned} \quad (38)$$

where the previous coefficients α_i are now replaced by functions, as follows:

$$F_{k,C} = \bar{c}_{k,i+1/2,j+1/2}^n - \frac{\lambda \bar{c}_{k,i+1/2,j+1/2}^n}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}} \times (a_{\phi,q,i,j+1/2} + a_{\phi,q,i+1,j+1/2} + a_{\xi,q,i+1/2,j} + a_{\xi,q,i+1/2,j+1} + a_{\phi,b,i,j+1/2} + a_{\phi,b,i+1,j+1/2} + a_{\xi,b,i+1/2,j} + a_{\xi,b,i+1/2,j+1}), \quad (39)$$

$$F_{k,N} = \frac{\lambda}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}} \times ([a_{\xi,q,i+1/2,j+1} + a_{\xi,q,i+1/2,j+1}]\bar{c}_{k,i+1/2,j+3/2}^n - \frac{\Psi_{i+1,j+1} - \Psi_{i,j+1}}{2} q_{k,0}(\bar{c}_{k,i+1/2,j+3/2}^n) - b[r_a H^3 \cos \beta]_{i+1/2,j+3/2} I_{k,3}(\bar{c}_{k,i+1/2,j+3/2}^n)), \quad (40)$$

$$F_{k,E} = \frac{\lambda}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}} \times ([a_{\phi,q,i+1,j+1/2} + a_{\phi,b,i+1,j+1/2}]\bar{c}_{k,i+3/2,j+1/2}^n + \frac{\Psi_{i+1,j+1} - \Psi_{i,j+1}}{2} q_{k,0}(\bar{c}_{k,i+3/2,j+1/2}^n) + b[r_a H^3 \sin \beta \sin \pi \phi]_{i+3/2,j+1/2} I_{k,3}(\bar{c}_{k,i+3/2,j+1/2}^n)), \quad (41)$$

$$F_{k,S} = \frac{\lambda}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}} \times ([a_{\xi,q,i+1/2,j} + a_{\xi,q,i+1/2,j}]\bar{c}_{k,i+1/2,j-1/2}^n + \frac{\Psi_{i+1,j} - \Psi_{i,j}}{2} q_{k,0}(\bar{c}_{k,i+1/2,j-1/2}^n) + b[r_a H^3 \cos \beta]_{i+1/2,j-1/2} I_{k,3}(\bar{c}_{k,i+1/2,j-1/2}^n)), \quad (42)$$

$$F_{k,W} = \frac{\lambda}{2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2}} \times ([a_{\phi,q,i,j+1/2} + a_{\phi,b,i,j+1/2}]\bar{c}_{k,i-1/2,j+1/2}^n - \frac{\Psi_{i,j+1} - \Psi_{i,j}}{2} q_{k,0}(\bar{c}_{k,i-1/2,j+1/2}^n) - b[r_a H^3 \sin \beta \sin \pi \phi]_{i-1/2,j+1/2} I_{k,3}(\bar{c}_{k,i-1/2,j+1/2}^n)). \quad (43)$$

Note that $q_{k,0}(0) = 0$ and $I_{k,3}(0) = 0$ for each k , so that $F_{k,l}(0) = 0$ for each $l = N, E, S, W$. Further, due to the selection of the various positive a -terms, we see that each $F_{k,l}$ is strictly increasing for $l = N, E, S, W$ and hence is positive. The function $F_{k,C} \geq 0$. We see that it is increasing for $\bar{c}_{k,i+1/2,j+1/2}^n \geq 0$, provided that

$$\Delta t \leq \Delta \phi \Delta \xi \min_{ij} \{\Lambda_{q,b,i,j}\}, \quad \text{where} \\ \Lambda_{q,b,i,j} = 2(\bar{H}\bar{r}_a)_{i+1/2,j+1/2} / [a_{\phi,q,i,j+1/2} + a_{\phi,q,i+1,j+1/2} + a_{\xi,q,i+1/2,j} + a_{\xi,q,i+1/2,j+1} + a_{\phi,b,i,j+1/2} + a_{\phi,b,i+1,j+1/2} + a_{\xi,b,i+1/2,j} + a_{\xi,b,i+1/2,j+1}]. \quad (44)$$

Inequality (44) is the CFL condition for the D2DGA discretization. It ensures that each $\bar{c}_{k,i+1/2,j+1/2}^{n+1} \geq 0$ provided that the surrounding values on the five-point stencil are positive at timestep n . We again calculate the timestep (Δt) by setting equality in (44) and multiplying the right-hand side of (44) by a fixed CFL number < 1 .

Assuming as before that $\sum_{k=1}^K \bar{c}_{k,i+1/2,j+1/2}^n = 1$, and each $\bar{c}_{k,i+1/2,j+1/2}^n \in [0, 1]$, consider summing each $F_{k,l}$ for $k = 1, \dots, K$. The terms multiplying any a -term sum to 1. The terms involving $I_{k,3}$ vanish. The terms involving $q_{k,0}$ also sum to 1. These latter multiply the differences in the stream function. When (38) is also summed over $k = 1, \dots, K$, the stream function terms cancel each other out. This leaves only the various positive a -terms which occur in pairs canceling with the identical negative terms in $F_{k,C}$. Consequently, we are left only with

$$\sum_{k=1}^K \bar{c}_{k,i+1/2,j+1/2}^{n+1} = \sum_{k=1}^K \bar{c}_{k,i+1/2,j+1/2}^n = 1.$$

It follows that each $\bar{c}_{k,i+1/2,j+1/2}^{n+1} \in [0, 1]$.

E. Numerical examples

We now repeat the displacement examples from earlier (cases 1–3, and 5), using the Newtonian D2DGA model (recall that case 4 involved non-Newtonian fluids). This helps demonstrate that the conservation properties of the numerical scheme developed above still hold for more complex fluxes and offers an initial comparison between the 2DGA and D2DGA models.

Figure 9 shows the results for case 1, which is steady stable displacement in a vertical low eccentricity annulus. This can be compared with Fig. 3 from earlier. The initial fluid 1 to fluid 2 front is again sharp, but unlike the 2DGA model, the flow retains a sharp contrast for small $\bar{c}_2$. We have again plotted black lines at $\bar{c}_1 = 0.9$, $\bar{c}_1 = 0.5$, $\bar{c}_1 = 0.1$ to illustrate the spreading of the displacement front. We see that the spreading is very asymmetric about

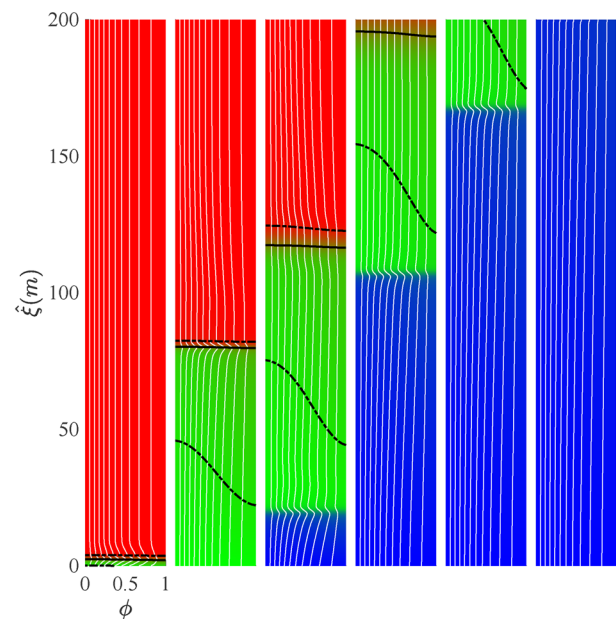


FIG. 9. D2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9$, and 2987.5 s are shown for case 1. White lines represent streamlines and black lines correspond to $\bar{c}_2 = 0.1, 0.5$, and 0.9 from top to bottom. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

$\bar{c}_1 = \bar{c}_2 = 0.5$, with significant dispersion behind and less ahead. This pattern is repeated for the fluid 2 to fluid 3 front. It is interesting to note first that the dispersive effects neglected in the 2DGA approach are indeed significant even for a simple flow. Second, considering the asymmetrical behavior observed, dispersion does not only increase spreading of the front but can also sharpen it. In this case, the dispersion has reduced the effects of numerical diffusion observed ahead of the front in Fig. 3.

Cases 2 and 3 are shown in Figs. 10 and 11, respectively. These differ considerably from their 2DGA counterparts in Figs. 12 and 13. For the horizontal displacement of case 2, the azimuthal component of the buoyancy flux, involving $I_{k,3}$, is active in dispersing the displacing fluid around to the narrow side of the annulus. Hence, surprisingly the dispersive displacement is much more effective than predicted by the 2DGA model. Fluid 3, also denser and more viscous, follows a similar behavior in displacing fluid 2.

The density unstable configuration of case 3 is also aided by dispersion. The very mild eccentricity of the annulus that led to a long finger of fluid 2 advancing along the wide side is insufficient to overcome dispersive fluxes in the D2DGA model. Although the fluid 1–fluid 2 front shows some evident instability, the long finger does not emerge and fluid 2 remains largely distributed around the annulus, as it disperses longitudinally. The denser and more viscous blue fluid 3 displaces very effectively, but also with more dispersion axially than predicted by the 2DGA model.

Figure 14 explores the impact of the dispersion on the slumping flow for case 5. Comparing Fig. 14 with Fig. 6, both 2DGA and D2DGA models generally predict a slumping front moving to the lower side of the annulus. However, the main discrepancy is associated with the degree of dispersion of the fronts, both ahead and behind those of the 2DGA model. As noted in case 2, the azimuthal buoyancy

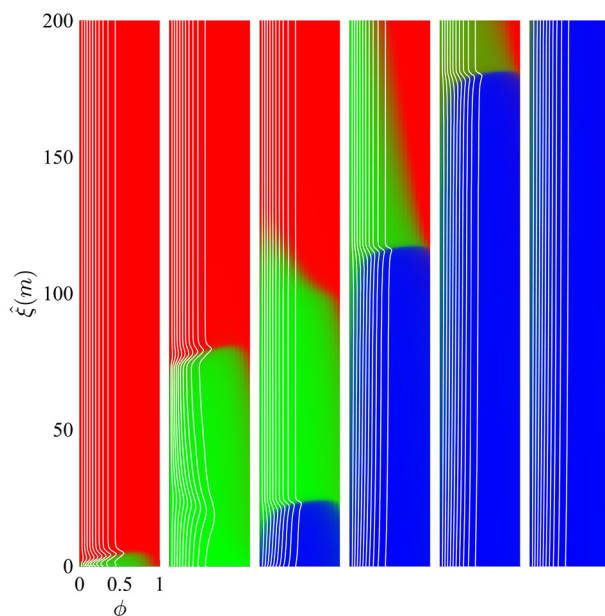


FIG. 10. D2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9$, and 2987.5 s are shown for case 2. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

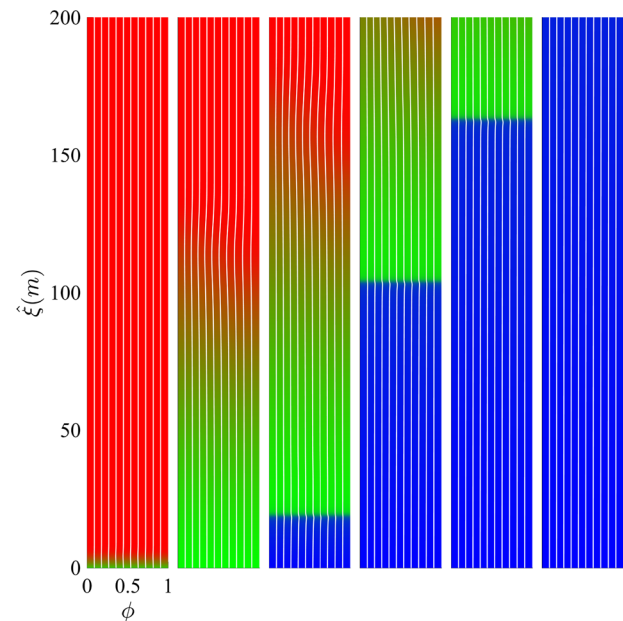


FIG. 11. D2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9$, and 2987.5 s are shown for case 3. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

fluxes augment transport. These then contribute to the increased axial dispersion that is evident. Thus, the displacing fluid exits the annulus earlier for D2DGA than for the 2DGA model (faster breakthrough times) (see Fig. 14).

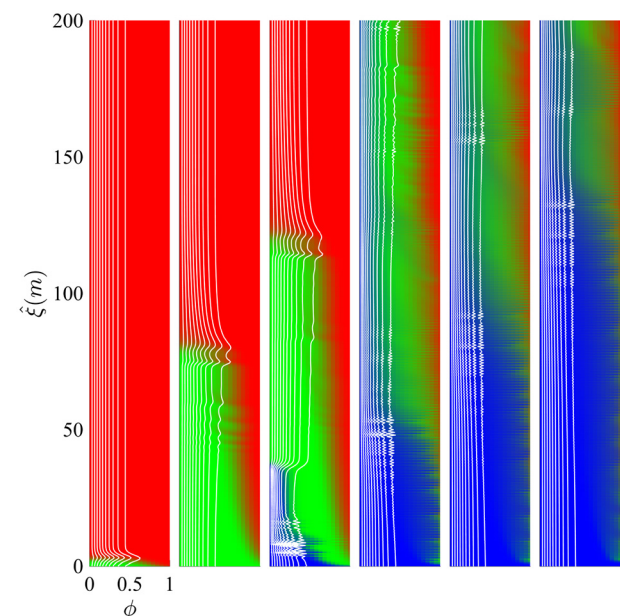


FIG. 12. 2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9$, and 2987.5 s are shown for case 2. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

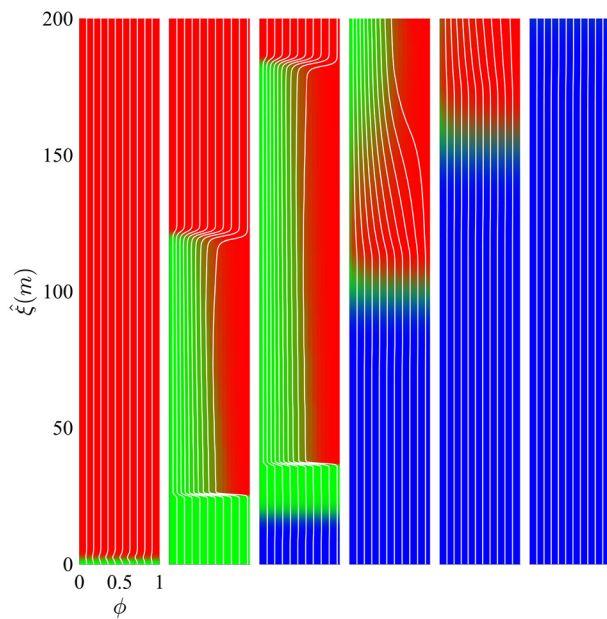


FIG. 13. 2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9,$ and 2987.5 s are shown for case 3. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

In Fig. 15, we show the relative volume errors for each of the three fluids, computed again for cases 1, 3, and 5, using the D2DGA model and numerical method outlined in the previous section. As we can see, the errors are comparable to those for the simpler 2DGA model and discretization.

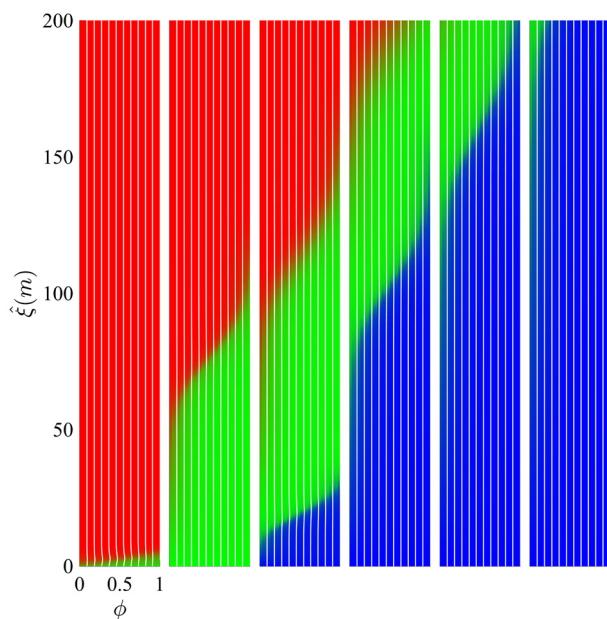


FIG. 14. D2DGA model: gap-averaged concentration color maps at $\hat{t} = 23.9, 717, 1075.5, 1864.2, 2413.9,$ and 2987.5 s are shown for case 5. White lines represent streamlines. Fluid 3 (blue) is injected into the annulus at $\hat{t} = 900$ s.

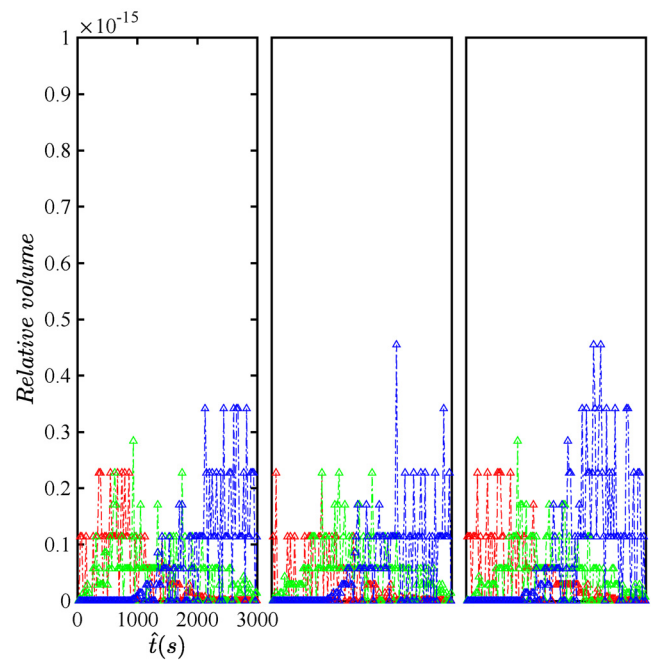


FIG. 15. D2DGA model: Conservation errors of fluid 1 (red), fluid 2 (green), and fluid 3 (blue) for cases 1, 3, and 5 (from left to right).

VI. DISCUSSION AND CONCLUSIONS

In this paper, we have studied the flows that occur with the industrial process of primary cementing. These are sequential displacement flows typically involving $K > 2$ shear-thinning yield stress fluids. The flows occur along extremely long, narrow annuli, formed by the inner steel casing and outer borehole wall (or previous casing). The geometric specificity of these flows (narrow annular gap), motivates the use of asymptotic methods/scaling arguments that reduce the Navier–Stokes equations to a two-dimensional (2D) shear flow and finally, to a 2D elliptic equation for the stream function of the gap-averaged velocity field. This is a classical approach used for Hele–Shaw cell flows, adapted to cementing displacement flows of both Newtonian and non-Newtonian fluids.¹² The stream function equation depends on the annular geometry (typically eccentric) and on the concentrations of fluids present in the annulus, which in turn determine the physical properties. Time dependency enters only via the concentrations locally present, and potentially changes in imposed flow rate. Each individual fluid concentration $\bar{c}_k$ satisfies a conservation equation of form (4).

Whereas equations such as (4) are routinely solved in scalar form, when we have a system of concentration equations, there are additional constraints to satisfy. First, since each $\bar{c}_k$ represents a volume fraction, we expect that $0 \leq \bar{c}_k \leq 1$ everywhere in the well. Second, the volume fractions must sum to 1, at each position. Methods able to satisfy both these constraints, for a sequence of $K > 2$ fluids are not available to our knowledge and this has been the main focus and motivation for the paper.

To derive such a method, we used a conservative scheme with monotone fluxes, based on the local Lax–Friedrichs fluxes. We concentrated initially on the simple 2DGA model^{12,16} and neglected diffusion/dispersion. By using the conservation properties of the stream

function Ψ , we were able to develop a method for which the concentrations all remain bounded and sum to 1 on each cell. The efficacy of the method has been illustrated with a number of examples of increasingly complex flows, including combinations of non-Newtonian fluids.

Furthermore, we compared the volumes pumped into (and exiting) the domain, with the integral of the fluid concentrations within the domain, throughout the transient computations. This comparison consistently showed a relative error of $\sim 10^{-15}$, indicating the strong conservative properties of the scheme. In the industrial context, an accurate fluid volume computation is a key aspect of the operation design. It is common to pump excess fluid to account for, e.g., hole irregularities,²⁵ and there can be significant detrimental effects of a low top of cement.²⁶ Thus, the numerical method proposed satisfies key requirements, scientifically and industrially.

In the later part of the paper, we have extended the numerical method to a wider range of volumetric flux functions, also indicating how (typically small) diffusive terms can fit within the same framework. The D2DGA model¹⁸ is particularly notable, modeling the strong dispersion that can occur on the scale of the annular gap. The dispersive fluxes are influenced strongly by both buoyancy and viscosity ratios. With some modifications of the underlying approach, we were able to devise a numerical scheme that was stable, with monotone fluxes, that kept each $0 \leq \bar{c}_k \leq 1$, everywhere in the well, and for which the $\bar{c}_k$ sum to 1. This method was illustrated using the same examples as for the 2DGA model, showing similar relative volume errors.

Although not the aim of this paper, comparisons of the illustrative examples suggest that the D2DGA model gives significantly different results, due to the modeling of dispersive fluxes. While this comparison has been made for vertical displacement flows,¹⁸ the results for horizontal wells are perhaps more significant. Here the explicit occurrence of an azimuthal buoyancy driven flux appears key. We intend to explore this further in a subsequent study and compare with experiments.²⁷

For most of the examples shown in the paper, there is observable numerical diffusion in the results. This stems from using the Lax–Friedrichs fluxes, which are first order. We have used local Lax–Friedrichs fluxes (LLF), i.e., the various wavespeed terms are locally calculated, although a global constraint is needed for the overall numerical stability (monotonicity of the numerical fluxes), which leads to the CFL conditions in the paper. In Sec. III A, we show an example of what happens if global Lax–Friedrichs fluxes are used. In the example, we consider displacement in a strongly eccentric annulus, which is very common in cementing. As the annular gap-size H decreases, the velocities decrease accordingly. The LLF captures this feature, but the results from the global LF diffuse unphysically around the annulus. Thus, the LLF approach is certainly sensible.

Given the above, the only remaining option to control the numerical diffusion is to increase the order of the method, but without affecting the conservation properties. We believe this can be done by again adapting the methods outlined in Ref. 21, at least to bring the method to second order. This is part of our future development. Our other main direction is in developing a 2D numerical solution for the fully non-Newtonian D2DGA models recently developed.²⁰

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

F. Bararpour: Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **M. Carrasco-Teja:** Investigation (equal); Methodology (equal); Project administration (equal); Software (supporting); Validation (supporting); Writing – review & editing (supporting). **I. Frigaard:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: LOCAL AND GLOBAL LF SCHEME FOR HIGH ECCENTRIC ANNULI

In this paper, we have used the local LF scheme throughout. More clearly, in determining the various wavespeeds a , we have

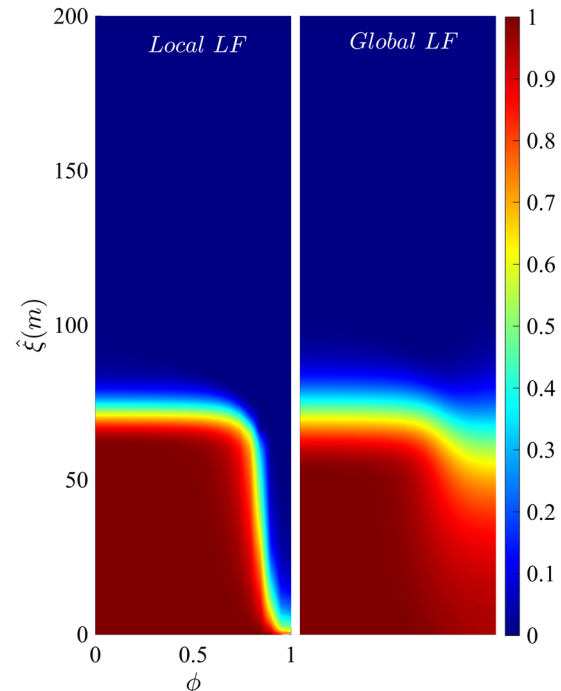


FIG. 16. 2DGA model: gap-averaged concentration color maps at $\hat{t} = 717$ s are shown for Newtonian fluid flows at $\hat{\rho}_1 = 1200 \text{ kg/m}^3$, $\hat{\rho}_2 = 1300 \text{ kg/m}^3$, $\hat{\kappa}_1 = 0.05 \text{ Pa s}$, $\hat{\kappa}_2 = 0.075 \text{ Pa s}$, $e = 0.9999$, $\beta = 0$.

used wavespeeds computed using the local values of the concentration. The largest wavespeeds are of course then used to ensure stability via the CFL condition. It might be questioned why we used the local LF instead of assigning a global maximum wavespeed value to each flux.

This would be possible to do and would not affect the monotonicity arguments we have made. However, the global value is too large for those parts of the annulus where the local wavespeed is small. For some problems, this might not be a problem, but in the primary cementing application, the annuli are often very eccentric, meaning that the velocities can be systematically small in large parts of the annulus, which are also those parts of the annulus that are of interest.

To illustrate the problems that arise, Fig. 16 presents results of a 2DGA displacement with very high eccentricity (and vertical). The LLF method correctly captures the slowly moving narrow side channel of residual fluid 1. On the other hand, the global LF method results in unrealistic numerical diffusion around the narrow side. An additional concern here can also be that, if not coded correctly, the timestep restriction can be severe, e.g., if calculated using the largest wavespeed and smallest gap size.

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